

# Mend-Amar Badral

 MendeBadra |  Mend-Amar Badral |  mendebadra.github.io  
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Driven by a problem solving mindset and fueled by curiosity, I aspire to develop genuine solutions for the real-world, impactful problems.

## SKILLS

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**Languages:** English (C1 [IELTS 8.0](#) )  
**Programming:** Python, SQL, Bash, Julia, C/C++, R  
**Tools:** Linux, Git, Docker, pandas, PyTorch, sklearn, opencv, open3d, Quarto, L<sup>A</sup>T<sub>E</sub>X

## EDUCATION

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**Budapest University of Technology and Economics (BME)** Sep, 2025 – June, 2027  
M.Sc. in Computer Science Engineering  
Budapest, Hungary  
*Supervisor:* [Márton Vaitkus](#)   
*Specialization:* Data Science & Artificial Intelligence

**National University of Mongolia (NUM)** Sep, 2021 – June, 2025  
B.Sc. in Applied Mathematics  
Ulaanbaatar, Mongolia  
GPA: 3.2/4.0 (**rank 3/34**)  
*Supervisor:* [Galtbayar Artbazar](#)   
*Thesis:* [Evaluating land degradation using image processing methods](#) 

## EXPERIENCE

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**Center of Mathematics Application, NUM** Oct, 2023 – June, 2025  
*Undergraduate Research Intern*

- **Data Processing & Python:** Developed custom [Python pipelines](#)  multi-threading to process large-scale datasets (1,800+ multispectral images). Optimized data ingestion and processing workflows, reducing execution time by 80% (5x factor).
- **Data Infrastructure:** Orchestrated photogrammetry pipelines using **OpenDroneMap** and **Docker** with **GPU acceleration** on Linux. Managed complex dependency environments and GPU debugging to ensure reliable high-throughput processing of spatial data.
- **Data Management:** Utilized **SQL** and relational database principles to organize research metadata and spatial results, facilitating efficient querying and data integrity across multiple field studies.

## PROJECTS

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### [aiSim Data Integration for 3D Gaussian Splatting](#)

- Developed a sensor fusion pipeline to parse and convert over sensor measurements from aiMotive's [aiSim](#)  ADAS simulation platform into the [nerfstudio](#)  3D scene representation format. Worked on the camera coordinate system transformation from sensor frame to the world frame.

### [Exploratory Data Analysis – Bike Sharing Dataset](#)

- Performed exploratory data analysis on bike sharing dataset. Published the results personal blog website using Quarto.

### [Deep Learning Foundations and Concepts Book Examples Reproduction](#)

- Reproduced polynomial regression experiments from Bishop & Bishop's *Deep Learning Foundations and Concepts* using Julia notebooks to using interactive sliders.

## AWARDS

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**Stipendium Hungaricum Scholarship** Sep, 2025